CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 16: Area Theorems

Topic-wise short notes with solved examples

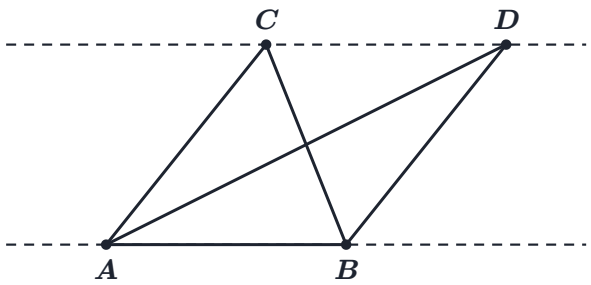
Theorems comparing the areas of parallelograms and triangles on the same base and between the same parallels, with the results on medians and their use in proofs.

1 Basic ideas

- **Congruent figures have equal areas.** Figures with equal areas need not be congruent.
- If a figure is divided into parts that do not overlap, its area is the sum of the areas of the parts.
- Two figures are **on the same base and between the same parallels** if they share a base and the vertices (or sides) opposite the base lie on one line parallel to the base.
- Figures between the same parallels have the **same height**.

2 The theorems

- **Theorem 1**: a parallelogram and a rectangle on the same base and between the same parallels are **equal in area**. So the area of a parallelogram $=$ base $\times$ height.
- **Theorem 2**: parallelograms on the same base (or equal bases) and between the same parallels are **equal in area**.
- **Theorem 3**: the area of a triangle is **half** the area of a parallelogram on the same base and between the same parallels. So the area of a triangle $= \frac{1}{2} \times$ base $\times$ height.
- **Theorem 4**: triangles on the same base (or equal bases) and between the same parallels are **equal in area**.
- **Converse of Theorem 4**: triangles with equal areas on the same base, and on the same side of it, lie between the same parallels.



Triangles ABC and ABD are on the same base AB and between the same parallels

Example 2.1 Parallelogram ABCD and rectangle ABEF are on the same base AB and have equal heights. If $AB = 8$ cm and the height is 5 cm, find the area of each.

SOLUTION

40 sq cm each

Example 2.2 P is a point on side DC of parallelogram ABCD, whose area is 72 sq cm. Find the area of triangle APB.

SOLUTION

The triangle and the parallelogram are on the same base AB and between the same parallels AB and DC.

36 sq cm

3 Medians and areas

- **A median divides a triangle into two triangles of equal area.** Both have equal bases and the same height.
- The three medians divide a triangle into **six triangles of equal area**.
- If G is the centroid of triangle ABC: area(GAB) $=$ area(GBC) $=$ area(GCA) $= \frac{1}{3}$ area(ABC).
- If D is a point on BC, then $\frac{\text{area}(ABD)}{\text{area}(ADC)} = \frac{BD}{DC}$, because the two triangles have the same height.

Example 3.1 AD is a median of triangle ABC and E is the mid-point of AD. If the area of triangle ABC is 48 sq cm, find the area of triangle BED.

SOLUTION

Area(ABD) $= 24$ sq cm. BE is a median of triangle ABD.

12 sq cm

Example 3.2 D is a point on side BC of triangle ABC with $BD : DC = 2 : 3$. If the area of triangle ABC is 60 sq cm, find the area of triangle ABD.

SOLUTION

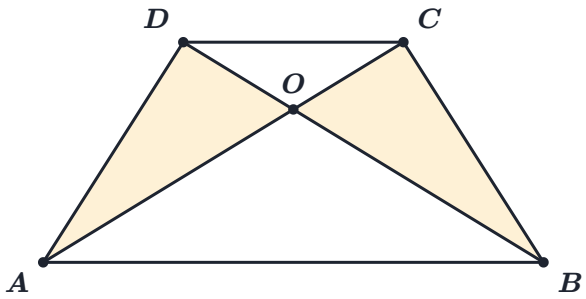
$\frac{2}{5} \times 60$

24 sq cm

4 Diagonals of quadrilaterals

- A diagonal of a parallelogram divides it into two triangles of equal area.
- The two diagonals of a parallelogram divide it into **four triangles of equal area**.
- In a **trapezium** ABCD with $AB \parallel DC$ and diagonals meeting at O: area(AOD) $=$ area(BOC).

- Area of a rhombus $= \frac{1}{2} \times d_1 \times d_2$.
- If the mid-points of the sides of a parallelogram are joined in order, the parallelogram formed has half the area.



Trapezium ABCD: the shaded triangles AOD and BOC have equal areas

Example 4.1 In trapezium ABCD, $AB \parallel DC$ and the diagonals meet at O. Prove that area(AOD) $=$ area(BOC).

SOLUTION

Triangles ADB and ACB are on the same base AB and between the same parallels, so their areas are equal.

Subtracting area(AOB) from both gives area(AOD) $=$ area(BOC)

Example 4.2 The diagonals of parallelogram ABCD meet at O. If the area of triangle AOB is 9 sq cm, find the area of the parallelogram.

SOLUTION

36 sq cm

5 Using the theorems in proofs

- Look for a common base and a pair of parallel lines.
- Add or subtract a common area to pass from one pair of equal areas to another.
- To prove two lines parallel, show that two triangles on the same base have equal areas (converse).

Example 5.1 In triangle ABC, D and E are points on AB and AC such that area(DBC) $=$ area(EBC). Prove that $DE \parallel BC$.

SOLUTION

Triangles DBC and EBC are on the same base BC, on the same side of it, and have equal areas.

They lie between the same parallels, so $DE \parallel BC$

Example 5.2 ABCD is a parallelogram and P is any point inside it. Prove that area(APB) + area(PCD) $= \frac{1}{2}$ area(ABCD).

SOLUTION

Through P draw a line parallel to AB, meeting AD at E and BC at F. Then ABFE and EFCD are parallelograms.

Area(APB) $= \frac{1}{2}$ area(ABFE) and area(PCD) $= \frac{1}{2}$ area(EFCD).

Adding gives $\frac{1}{2}$ area(ABCD)

Example 5.3 D, E and F are the mid-points of sides BC, CA and AB of triangle ABC. Prove that area(DEF) $= \frac{1}{4}$ area(ABC).

SOLUTION

By the mid-point theorem, BDEF, DCEF and AFDE are parallelograms. A diagonal divides a parallelogram into two equal triangles.

So triangles AFE, BDF, DCE and DEF all have equal areas.

Each is $\frac{1}{4}$ of area(ABC)

Example 5.4 In quadrilateral ABCD, a line through D parallel to AC meets BC produced at E. Prove that area(ABE) $=$ area(ABCD).

SOLUTION

Triangles ACD and ACE are on the same base AC and between the parallels AC and DE, so their areas are equal.

Adding area(ABC) to both: area(ABCD) $=$ area(ABE)

6 Common mistakes

- Using a theorem when the figures are between the same parallels but **not** on the same or equal bases.
- Forgetting the factor $\frac{1}{2}$ between a triangle and a parallelogram.
- Assuming equal areas mean congruent triangles.
- Using a slant side as the height.
- In a trapezium, claiming that triangles AOB and COD have equal areas. It is AOD and BOC that are equal.